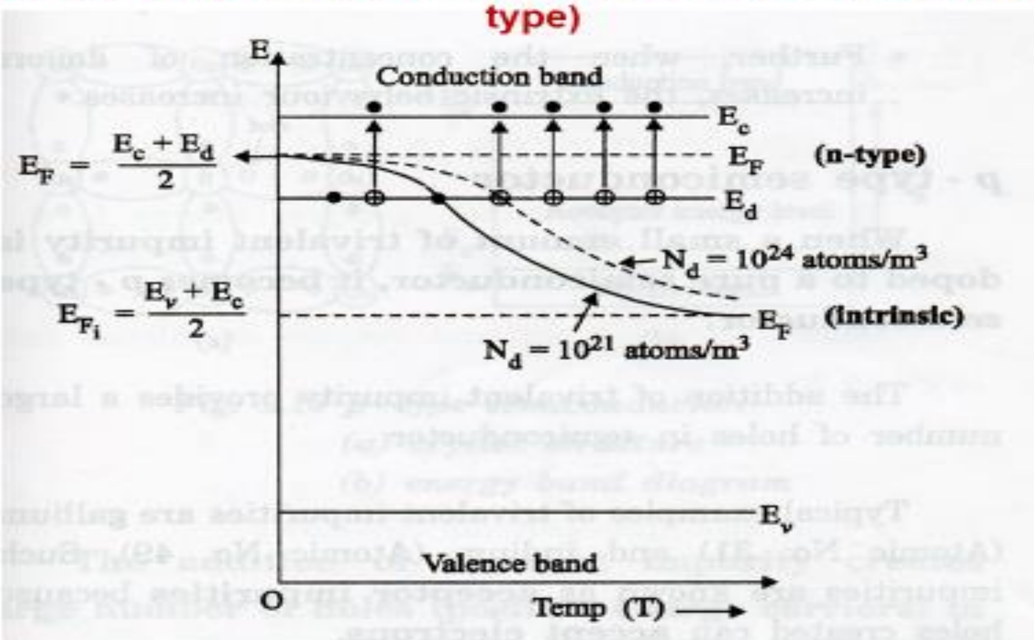
22PH202 – ELECTROMAGNETISM AND MODERN PHYSICS

Unit-V: ENERGY BANDS IN SOLIDS_LM5

Variation of Fermi Level with Temperature and Impurity (n-type)

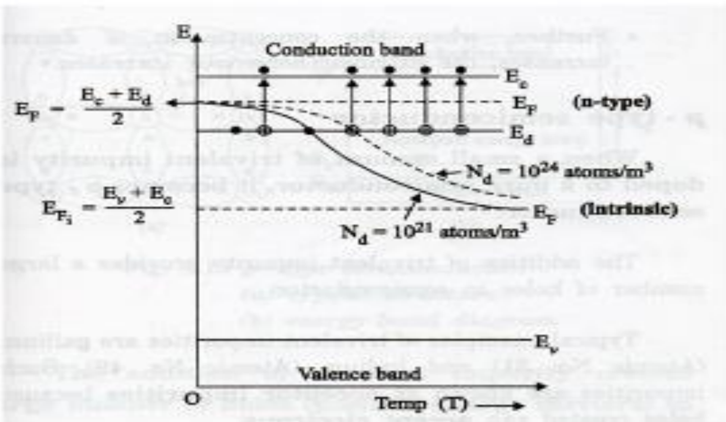


Variation of Fermi Level with Temperature and Impurity (n-type)

At $T = 0\,K$

$$E_F = \frac{E_C + E_d}{2}$$

The Fermi level lies exactly in the middle of the donor energy level and bottom of the conduction band



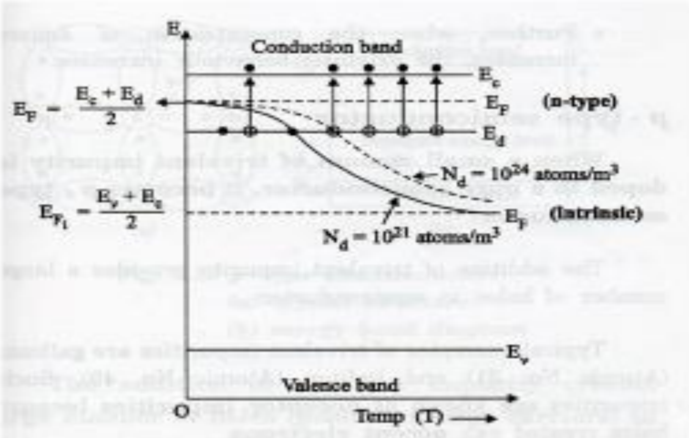
Variation of Fermi Level with Temperature and Impurity (n-type)

When the temperature is increased gradually, the contribution of electrons to the conduction band from the valence band increases

At very high temperatures (500 K), concentration of electrons in the conduction band due to breaking of covalent band (from valence band) is higher than the contribution from donor atoms

(Intrinsic behaviour dominates)

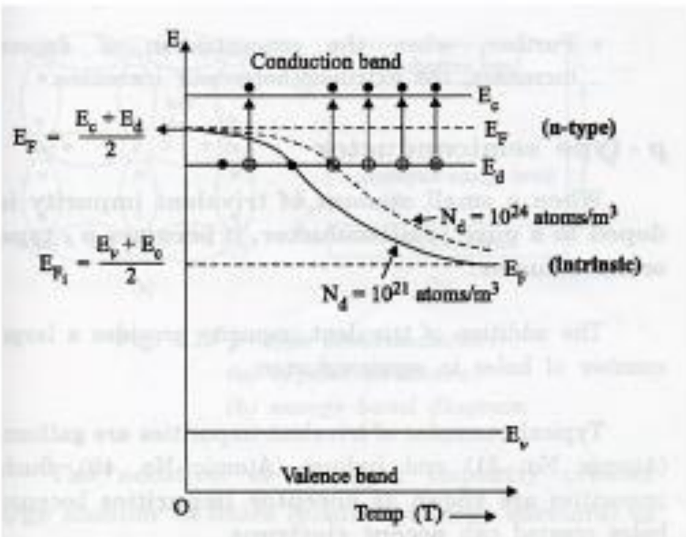
Fermi level shifts downwards and reaches the middle of the band gap



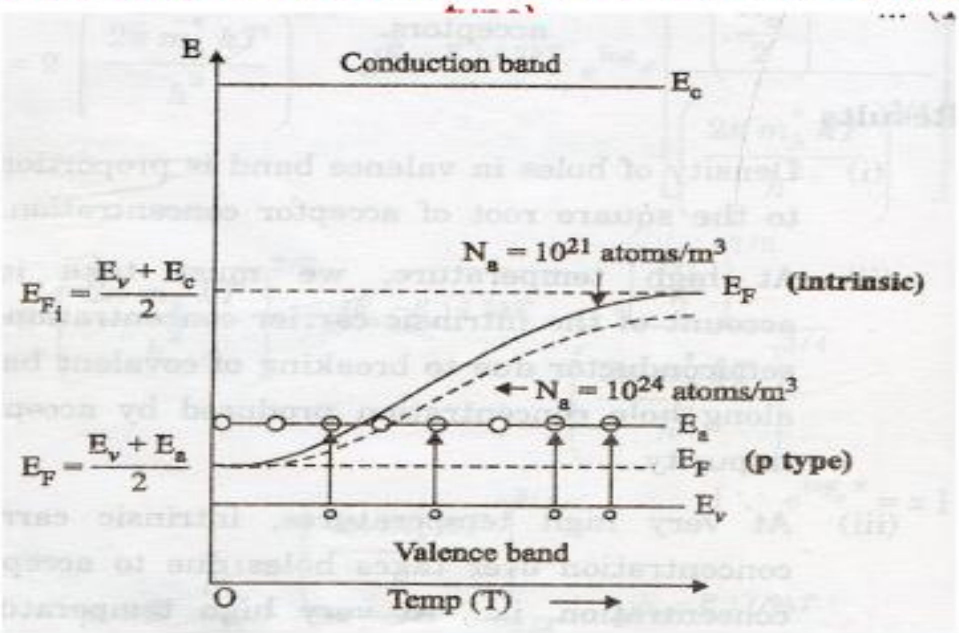
Variation of Fermi Level with Temperature and Impurity (n-type)

When the concentration of donors increases, the extrinsic behavior increases

The Fermi level shifts slightly upwards



Variation of Fermi Level with Temperature and Impurity (p-type)

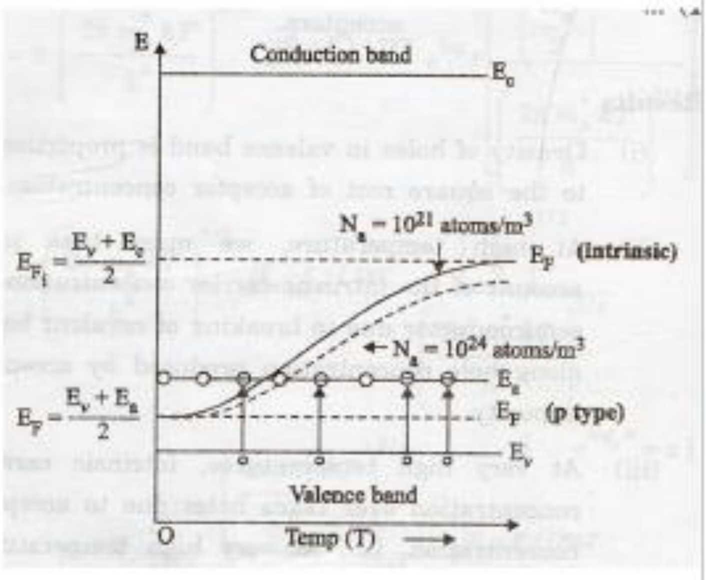


Variation of Fermi Level with Temperature and Impurity (p-type)

At T = 0 K

$$E_F = \frac{E_V + E_a}{2}$$

The Fermi level lies exactly in the middle of the acceptor energy level and top of the valence band



Variation of Fermi Level with Temperature and Impurity (p-type)

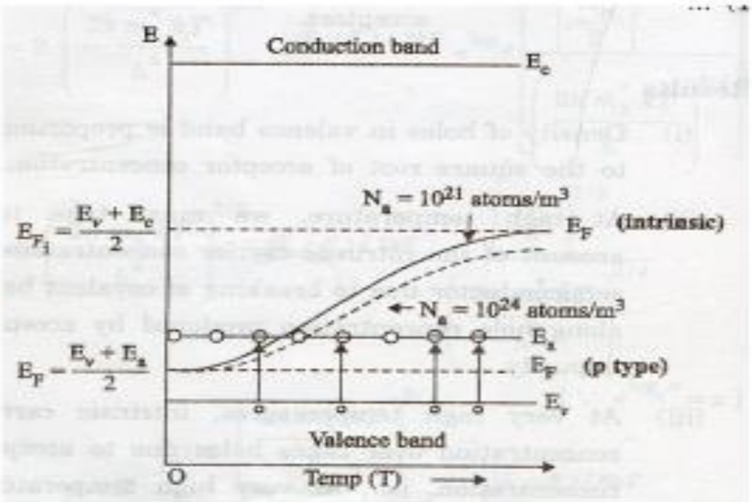
As temperature increases, more and more acceptor atoms are ionized

Fermi level shifts upwards

At a particular temperature, when all the acceptor atoms are ionized, Fermi level crosses acceptor level

At very high temperatures, Fermi level shifts to intrinsic Fermi level (Intrinsic behaviour dominates)

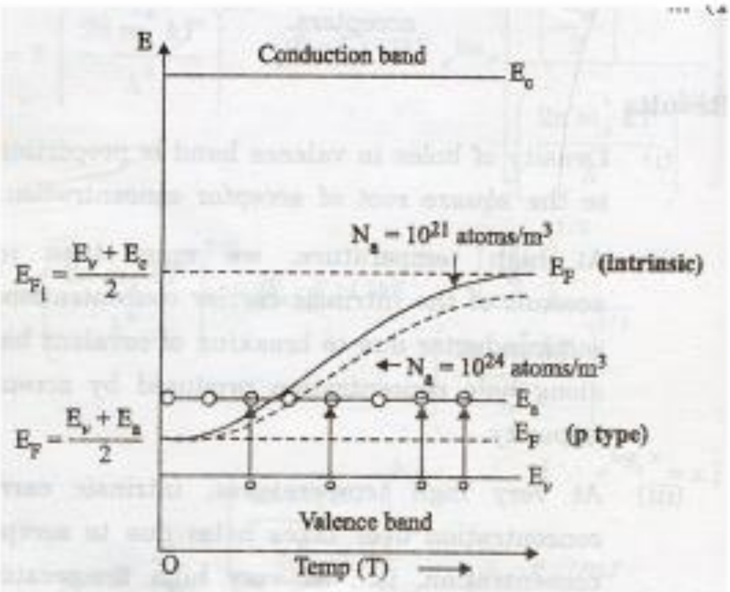
Fermi level shifts upwards and reaches the middle of the band gap

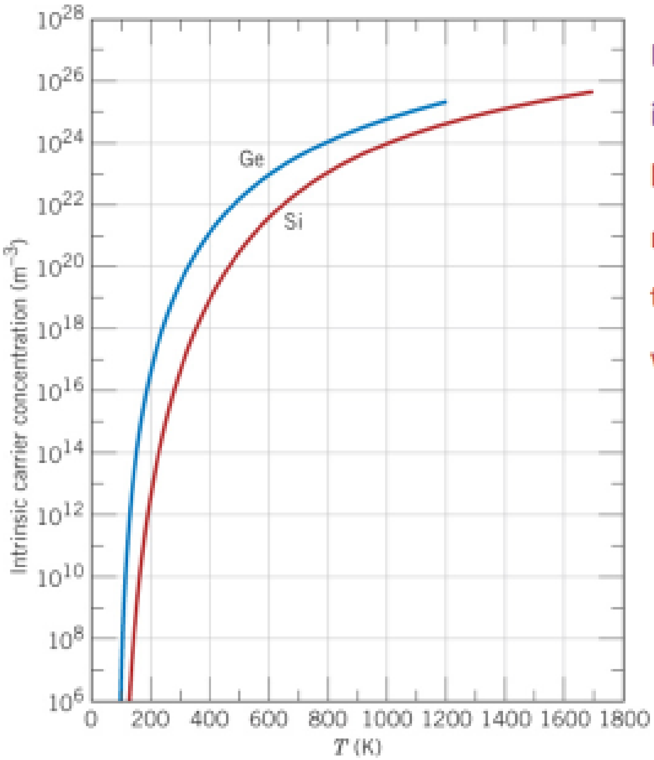


Variation of Fermi Level with Temperature and Impurity (p-type)

When the concentration of acceptor atoms increases, the extrinsic behavior increases

The Fermi level shifts slightly downwards





Electron and Hole concentrations increase with temperature because, with rising temperature more thermal energy is available to excite electrons from the valence band to conduction band

Electrical Conductivity – Intrinsic Semiconductors

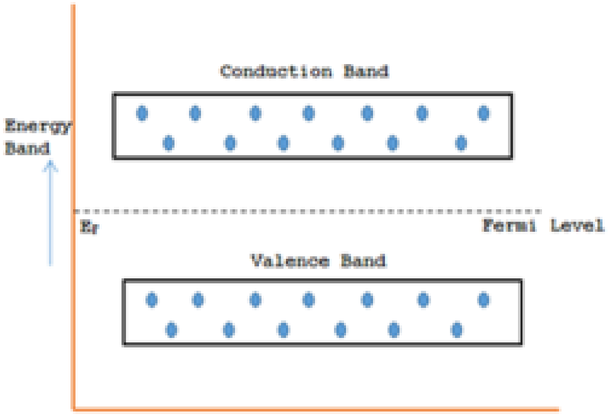
Intrinsic Carrier Concentration

$$n_i = n_e = n_h$$

$$n_i^2 = n_e \cdot n_h$$

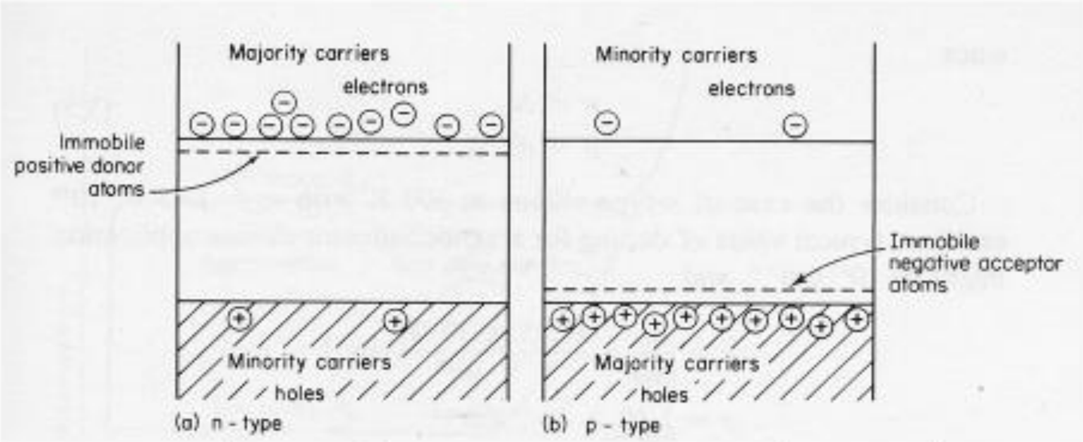
Electrical Conductivity

$$\sigma_i = n_i\, e\, (\mu_e + \mu_h)$$



Intrinsic semiconductor - Energy and Diagram

Electrical Conductivity – Extrinsic Semiconductors



$$\sigma = n_e\, e\left(\mu_e + \mu_h\right)$$

$$n_e >> n_h\ (\mu_h = 0)$$

$$\sigma = n_e\, e\ \mu_e$$

$$\sigma = n_p\, e\left(\mu_e + \mu_h\right)$$

$$n_h >> n_e\ (\mu_e = 0)$$

$$\sigma = n_h\, e\ \mu_h$$

Electrical Conductivity – Extrinsic Semiconductors

$$\sigma_i = n_e\, e\ \mu_e \qquad \sigma = n_h\, e\ \mu_h$$

Conductivity measurements in Semiconductors

Can not determine whether the conduction is due to electrons (n-type) or holes (p-type)

Not possible to distinguish between n-type and p-type semiconductors

Hall Effect Measurement in Semiconductors (Applications)

Type of charge carriers

Carrier concentration

Mobility